	Scotch Student ID #			
Circle the relevant digits	0	0	0	0
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	4	4	4	4
	5	5	5	5
	6	6	6	6
	7	7	7	7
	8	8	8	8
	9	9	9	9

Teacher's Name

Scotch College

CHEMISTRY

SAC 2 – RATES, EQUILIBRIUM & REDOX

MARKING SCHEME

Date: Wednesday, May 26th

Reading Time	0 minutes
Writing Time	45 minutes

Task Sections	Marks	Your Marks
Total Marks	41	

MC Answers	V1	V2	V3	V4	V5
Q1	A	A	B	A	B
Q2	B	A	D	C	C
Q3	D	A	D	C	C

Q4. *(9 marks)*

A mixture of 1.20 mol sulfur dioxide, SO_2 (g), 2.40 mol oxygen gas, O_2 (g) and 1.20 mol sulfur trioxide, SO_3 (g) is placed in a 1.00 L container and allowed to reach equilibrium at temperature T.

$$2SO_2\,(g) + O_2\,(g) \rightleftharpoons 2SO_3\,(g)\ \Delta H = -296.8\ \mathrm{kJ/mol}$$

a. Distinguish between the terms reaction quotient (concentration fraction), *Q,* and equilibrium constant, Kc. (2 marks)

WTTE: Reaction quotient refers to ratios of concentrations of products over reactants at a given time.*

Kc refers to rations of concentrations of products over reactants when concentrations are constant.*

b. The equilibrium constant, Kc, is 0.382 M^{-1} at temperature T. Deduce, showing your work, the direction (back or forward) of the initial reaction. (2 marks)

V1	V2	V3	V4	V5
Q = 0.416	Q = 0.445	Q = 0.500	Q = 0.435	Q = 0.304
Q>Kc*	Q>Kc*	Q>Kc*	Q>Kc*	Q<Kc*
Net back*	Net back*	Net back*	Net back*	Net forward*

c. Another equilibrium mixture was found to contain concentrations of 0.140 M SO_2 and 0.180 M oxygen gas, O_2 when the Kc is 1.75 M^{-1}. Calculate the equilibrium concentration of SO_3. (2 marks)

$$2SO_2\,(g) + O_2\,(g) \rightleftharpoons 2SO_3\,(g)\ \Delta H = -296.8\ \mathrm{kJ/mol}$$

V1	V2	V3	V4	V5
$\frac{(x)^2}{(0.140)^2(0.180)} = 1.75$ *	$\frac{(x)^2}{(0.110)^2(0.155)} = 1.75$ *	$\frac{(x)^2}{(0.120)^2(0.160)} = 1.75$*	$\frac{(x)^2}{(0.170)^2(0.185)} = 1.75$*	$\frac{(x)^2}{(0.165)^2(0.185)} = 1.75$ *
$[SO_3]$ = 0.0786 M *	$[SO_3]$ = 0.0573 M *	$[SO_3]$ = 0.0635 M *	$[SO_3]$ = 0.0967 M *	$[SO_3]$ = 0.0939 M *

- 1 mark if unit not included with final answer

d. Predict how each of the following changes affects the position of equilibrium and the value of Kc. You may use arrows, → ← ↑ ↓ (or 'no effect') to show your answer. (3 marks)

LCP answers	V1	V2	V3	V4	V5
Part i	NE, NE	Left, NE	Right, Up	Left, NE	NE, NE
Part ii	Right, NE	Left, Down	Right, NE	Left, Down	Right, Up
Part iii	Right, Up	NE, NE	NE, NE	NE, NE	Left, NE

Q5. *(3 marks)*

Alcohols can be oxidized with acidified potassium dichromate solution to form a variety of compounds including carboxylic acids. The balanced overall equation for one of these reactions is:

$$3CH_3OH\ (l) + 2Cr_2O_7^{2-}\ (aq) + 16H^+\ (aq) \rightleftharpoons 3CHOOH\ (aq) + 4Cr^{3+}\ (aq) + 11H_2O\ (l)$$

a. Identify the oxidising agent. Give your reasoning using oxidation numbers. (2 marks)

$Cr_2O_7^{2-}$ *;

Cr from +6 to +3 *

b. Deduce the reduction half-equation for this reaction. States are not required. (1 mark)

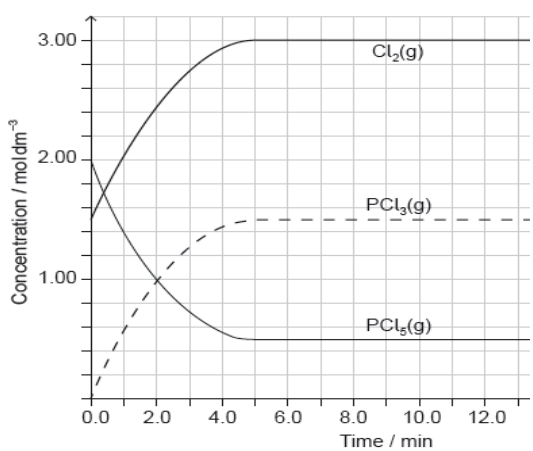
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$ *

Q4. (10 marks)

PCl_5 (g) decomposes according to the following equation:

$$PCl_5\ (g) \rightleftharpoons PCl_3\ (g) + Cl_2\ (g) \quad \Delta H = +87.8\ \text{kJ/mol}$$

PCl_5 (g) and Cl_2 (g) were placed in a sealed flask and allowed to reach equilibrium at 200 °C.



[Source: http://education.alberta.ca/media]

a. Calculate the value of the equilibrium constant, Kc. (2 marks)

[3.00] [1.50]/ [0.50] *

Kc = 9.0 M* (consequential if values in expression incorrect)

b. For the following **two** characteristics of a dynamic equilibrium, fill in the blank left after each statement to make them correct. (2 marks)

When in a state of dynamic equilibrium:

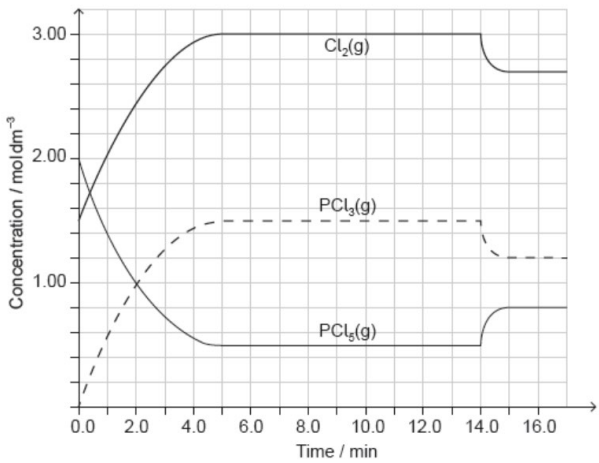
i. the rates of the forward and back reactions will

_____be equal *_____.

ii. the concentrations of products and reactants will

_____be constant *_____.

$$PCl_5\,(g) \rightleftharpoons PCl_3\,(g) + Cl_2\,(g) \quad \Delta H = +87.8\ \mathrm{kJ/mol}$$



[Source: http://education.alberta.ca/media]

c. Identify the **specific** change responsible for the equilibrium shift observed after 14 minutes. Explain your answer. (2 mark)

Products are increasing and reactants decreasing *

This happens when temperature decreases for an endothermic reaction *

d. How does each of the following changes affect the amount of PCl_3 in the equilibrium mixture? In your answers, refer to Le Chatelier's principle. (4 marks)

WTTE: [x] so LCP states that the system will partially oppose the change to [opposite to x] * by favoring the net [y] (1/2) and [z] yield (1/2).

LCP answers	V1	V2	V3	V4	V5
Part i	Increase P, net forward, increase yield	Increase P, net forward, increase yield	Increase P, net forward, increase yield	Decrease P, net back, decrease yield	Decrease P, net back, decrease yield

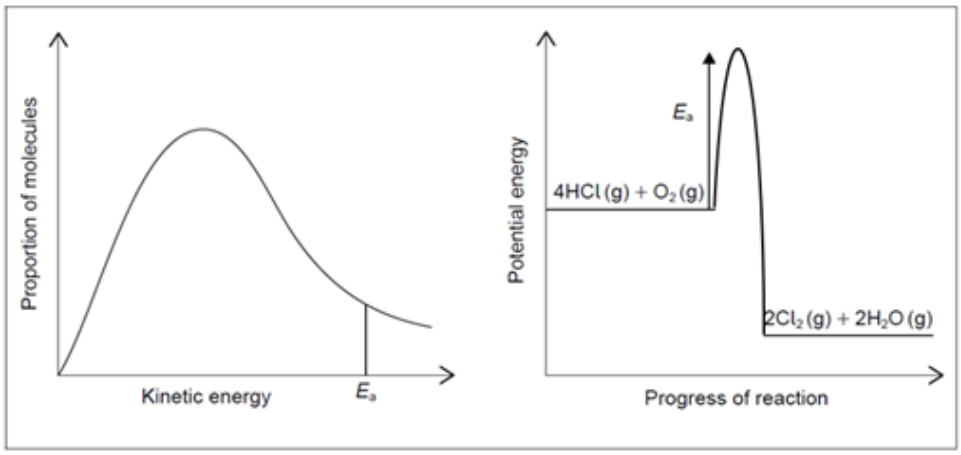
Part ii	Decrease [PCl_5], net forward, increase yield	Increase [PCl_3], net forward, increase yield	Increase [PCl_5], net back, decrease yield	Increase [Cl_2], net forward, increase yield	Increase [Cl_2], net forward, increase yield

Q5. (12 marks)

The diagram shows the Maxwell–Boltzmann distribution and potential energy profile for the following reaction without a catalyst.

$$4HCl\ (g) + O_2\ (g) \rightleftharpoons 2Cl_2\ (g) + 2H_2O\ (g)$$

a. Annotate **both** charts to show the activation energy for the catalysed reaction, using the label $\boldsymbol{E_{a\ (cat)}}$.

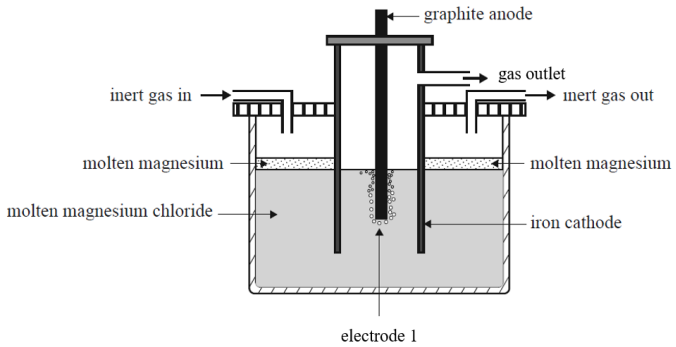


(2 marks)

b. Explain in general terms how the catalyst increases the rate of the reaction. (1 mark)

Catalyst lowers the activation energy (1/2) by providing an alternate pathway (1/2).

An electrolysis cell was assembled using a graphite electrode and an iron electrode and connected as shown. Inert gas flows through the cell to keep the pressure constant within the cell.



c. Write the half-equation for the formation of gas bubbles at electrode 1. (2 marks)

$2Cl^-$ (l) → Cl_2 (g) + $2e^-$ * Correct states *

d. Explain why the graphite electrode cannot be replaced with another iron electrode. (2 marks)

Fe (s) is a stronger reducing agent than Cl^-(l). *

If used, there would be side reactions of Cl_2 (g) with Fe (s) or Mg^{2+} (l) with Fe (s) *
(one example needed for second mark)

Fe would react preferentially to Cl^- and so cell would not operate
Fe would react preferentially and therefore dissolve

e. Using the electrochemical series, determine the applied voltage required for the electrolysis cell to operate. (1 mark)

> 3.73 *

f. If the cell ran for 2.00 minutes with a current of 1.20 A, calculate the volume of gas that would form at SLC. (4 marks)

V1	V2	V3	V4	V5
Q = 144 C *	Q = 270 C*	Q = 216 C *	Q = 210 C *	Q = 276 C *
n(e) = 1.49 x10^{-3} mol*	n(e) = 2.80 x10^{-3} mol*	n(e) = 2.24 x10^{-3} mol*	n(e) = 2.18 x10^{-3} mol*	n(e) = 2.86 x10^{-3} mol*
n(Cl_2) = 7.46 x10^{-4} mol*	n(Cl_2) = 1.40 x10^{-4} mol*	n(Cl_2) = 1.12 x10^{-3} mol*	n(Cl_2) = 1.09 x10^{-3} mol*	n(Cl_2) = 1.43 x10^{-3} mol*
V(Cl_2) = 1.85 x10^{-2} L*	V(Cl_2) = 3.47 x10^{-2} L*	V(Cl_2) = 2.78 x10^{-2} L*	V(Cl_2) = 2.70 x10^{-2} L*	V(Cl_2) = 3.55x10^{-2} L*

- 1 if final answer is not 3 significant figures.

Q6. (4 marks)

A reactivity series can be experimentally determined by adding the metals W, X, Y and Z to solutions of these metal ions. The following spontaneous reactions were observed:

$$W^{2+}(aq) + Z(s) \rightarrow W(s) + Z^{2+}(aq)$$

$$Y(s) + W^{2+}(aq) \rightarrow Y^{2+}(aq) + W(s)$$

$$Z^{2+}(aq) + Y(s) \rightarrow Z(s) + Y^{2+}(aq)$$

$$Y(s) + X^{2+}(aq) \rightarrow Y^{2+}(aq) + X(s)$$

a. Deduce the order of reactivity of these four metals, from the weakest reductant to the strongest reductant. (1 mark)

V1	V2	V3	V4	V5
W<Y<Z<X *	X<Y<W<Z *	Z<W<X<Y *	Y<W<Z<X *	Z<W<Y<X*

b. A secondary cell is made by connecting a half-cell of X in X^{2+} (aq) to a half-cell of Z in Z^{2+} (aq). Deduce the overall equation for the reaction taking place when the cell is **discharging**. (1 mark)

V1	V2	V3	V4	V5
$W^{2+} + Z \rightarrow W + Z^{2+}$ *	$X^{2+} + Z \rightarrow X + Z^{2+}$ *	$Z^{2+} + X \rightarrow Z + X^{2+}$ *	$Z^{2+} + X \rightarrow Z + X^{2+}$ *	$Z^{2+} + X \rightarrow Z + X^{2+}$ *

c. One of these cells was constructed and kept next to a heat source. Describe how this might affect the performance of this secondary cell over its lifetime. (2 marks)

High temperatures can lead to more side reactions in the cell *

This would decrease the battery performance over time *

End of SAC